

Joshua Vinu Koshy

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Education

Fr. C. Rodrigues Institute of Technology (FCRIT)

Navi Mumbai, India

Bachelor of Engineering in Computer Engineering | GPA: 9.74/10

2022 - 2026

Experience

Project Intern | InfinityPool Finnotech Pvt. Ltd. - [Letter of Completion](#) Jul 2024 – April 2026

- Designed a **Backtrader-powered trading engine** with multi-timeframe simulations, optimized data loaders, and real-time market feeds, enabling faster, production-ready strategy evaluation for equity portfolios.
- Architected **agentic AI workflows** with multi-agent orchestration and long-term memory, enabling autonomous equity monitoring and portfolio optimization for live portfolios.

Full Stack Developer Intern | FCRIT - [Letter of Completion](#)

Jan 2025 – Jun 2025

- Led a complete revamp of fcrit.ac.in with **React.js** UI redesign, admin dashboards, and backend migration from CakePHP to a **Node.js + MySQL** microservice-style architecture, reducing load time by 40%.
- Implemented secure REST APIs with JWT auth, RBAC, and caching for exams, faculty management, and feedback modules, improving reliability and maintainability of the portal.

Projects

Local LLM Powered Policy Gap Analysis and Improvement Module | Hack IITK 2026
Python, Mistral, Phi, Ollama, NLP, ChromaDB - [GitHub](#)

- Developed a **fully offline policy gap analysis system** using local LLMs (Phi & Mistral) deployed via Ollama, adhering to zero API and no-cloud constraints.
- Built a **RAG- based pipeline to analyze organizational policies** against NIST cybersecurity framework templates, identifying policy gaps by leveraging **vector embeddings & semantic search**.
- Implemented **Auto, Deep, and Express modes** to balance accuracy and latency & **generate improved policy** and actionable recommendations aligned with industry standards.

Financial Assistant using Agentic Workflows & Long-Term Memory | Jul 2025 – Apr 2026
Python, n8n, Next.js, GCP, Cloud SQL, Node.js, Selenium, LLMs

- Built a **production-grade financial assistant** using **agentic workflows (n8n)** and **LLM-based reasoning**, generating trading signals across **7,500+ equities** via **automated data pipelines**.
- Designed and orchestrated multi-agent architecture — **Fundamental, Valuation, Sentiment Agents** — enabling **data-driven decision making** with **structured + unstructured data integration**.
- Developed modular tools (**Stock Screener, Portfolio Analyzer, Chart Analyzer**) supporting **portfolio optimization, technical analysis, and end-to-end financial insights**.

Signal Agnostic Backtesting Framework |

Jul 2024 - Jun 2025

React.js, Node.js, PostgreSQL, Python, Backtrader, Selenium

- Engineered a **modular, event-driven backtesting engine** supporting **multi-strategy execution** with 15+ indicators, enabling **systematic trading strategy evaluation**.
- Built **high-performance data pipelines** with chunked **OHLCV time-series data ingestion**, memory mapping, and async caching, improving throughput by 35–40% & integrated quantitative performance metrics such as **Sharpe ratio, drawdown, volatility, equity curves** for risk-adjusted analysis.

Technical Skills

Languages: C/C++, JavaScript, Java, Python

Frameworks: Spring Boot, React.js, Next.js, Vite, Node.js, Express.js, Flask, HTML, CSS

Databases: MongoDB, MySQL, Firebase, PostgreSQL

AI/ML & LLMs: Generative AI (LLM-based), Machine Learning, AI/ML Integration

DevOps & Deployment: AWS (EC2, S3, Lambda), Docker, Google Cloud, Netlify, Vercel

Tools & Methodologies: Git, GitHub, Postman, JIRA, Google Colab, Agile, REST APIs

Leadership & Voluntary Experience

- Head, AI & Deep Learning Club** – Organized 10+ technical events for 250+ participants on campus.
- Speaker** for AI/ML workshop at **Cryptex 2024**, trained 30+ students.
- Guest speaker** at Short Term Training Program (Dept. of IT), delivered AI/ML sessions for 100+ attendees.

Publications & Achievements

- Published "[Fraudulent Transaction Detection in Reverse Auction Systems Using Machine Learning and Blockchain-Assisted Anomaly Analysis](#)," in International Conference on Nascent Technologies in Engineering 2026.
- 3rd place - Hack IITK 2026**, Cybersecurity hackathon by **C3iHub, IIT Kanpur** among **9000+ participants**
- 3rd Position** at **TechSprint 2026** (Open Innovation Hackathon) by **Google Developer Group, GSSSIETW**
- Second Runner-Up** at **Hacktiv8 2024** (AI + Cybersecurity hackathon), organized by Hacktify.
- Secured **first rank** in the third year of Computer Engineering with a perfect **GPA of 10.0**.
- 1st Prize** at **Cryptex 2023** (Web Development) & **3rd Prize** at **CodeMania**, a competitive programming contest organized by the **Computer Society of India**.